

Qatar-1b

R-0278

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_171104 · created 2026-07-17T14:43:39+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458061.6757 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.7 +/- 2.14 [%]
Orbital Inclination (inc)	88.7 +/- 2.83
Airmass coefficient 1 (a1)	1.34 +/- 0.26
Airmass coefficient 2 (a2)	-0.23 +/- 0.15
Scatter in the residuals of the lightcurve fit is	17.87 %
Best Comparison Star	#2 - [568, 77]
Optimal Aperture	2.43
Optimal Annulus	12.07
Transit Duration (day)	0.0827 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.1463 (deviation 0.73σ)
Duration fit vs published	0.0827 d vs 0.0692 d (deviation 1.57σ)
Mid-time O–C	1.6 min
Depth SNR	3.1
Residual scatter	17.87 %

Light curve

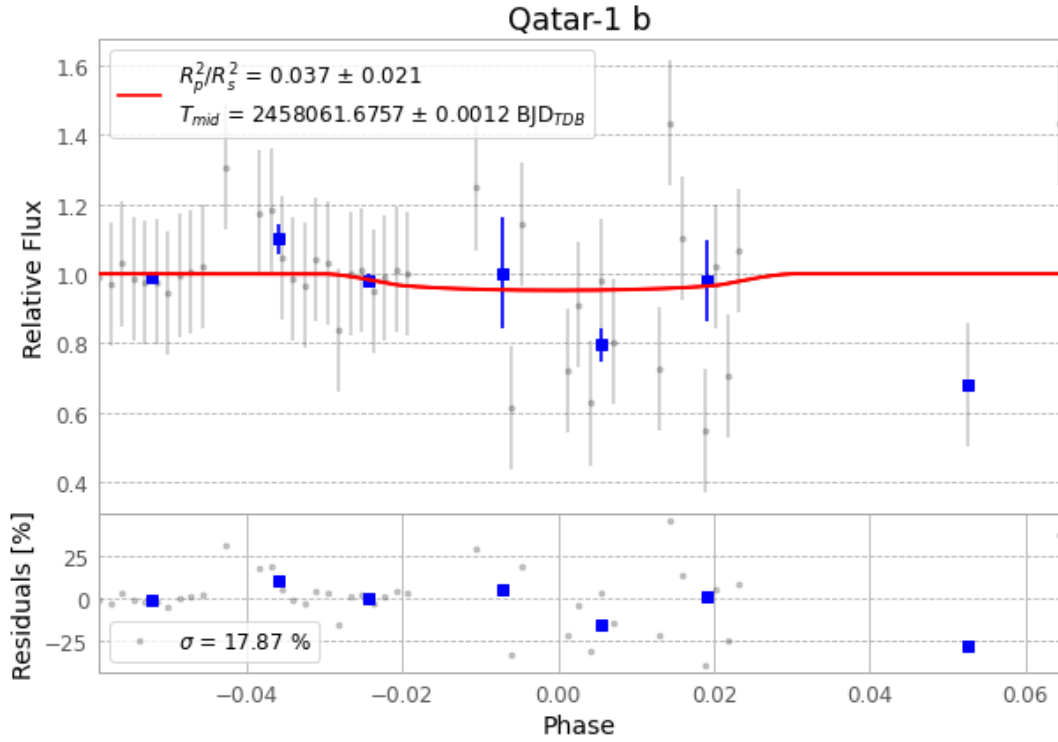


Figure 1 — FinalLightCurve_Qatar-1 b_2017-11-03.png (SHA-256 005c387be51fd96f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 349], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e92197b2f1bf546...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984

Data provenance

86 FITS frames (56 MB), Qatar-1171104020912.FITS → Qatar-1171104062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-171104.log (a71e3b1302c6...), FinalLightCurve_Qatar-1 b_2017-11-03.png (005c387be51f...), FinalLightCurve_Qatar-1 b_2017-11-03.csv (9b2440c8ba46...)

Compute resources

Wall time 270.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 14:43Z.